Fluid Universe Model: Full Dimensional Consistency Verification of Fundamental Physical Constants

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Abstract

Based on the fluid universe model (spacetime is a compressible cosmic superfluid with shear and bulk viscosity, and matter is a steady topological vortex soliton in the fluid), this paper takes only two closed-loop verified elementary parameters of a single vortex as the sole starting point: the elementary energy of a single vortex and the equivalent rest mass of a single vortex. Adhering to the three principles of no circular reasoning, no additional assumptions, and no fitting to match experimental values, we complete the first-principles derivation and full dimensional consistency verification of the full system of fundamental physical constants of physics along the unidirectional logical chain from fundamental constants to derived constants. This work is expected to provide a new path for quantum gravity and the standard model.

Keywords: Fluid Universe Model; Fundamental Physical Constants; First Principles; Dimensional Verification; Superfluid Spacetime; Topological Vortex

1 Introduction

The current Standard Model of particle physics contains more than 20 free fitting parameters, and the gravitational constant in general relativity is an independent empirical parameter. The two theories cannot be unified, and neither can provide an underlying dynamic origin explanation for these constants [2]. Meanwhile, existing theories have never completed the dimensional verification of the full system of constants, and the potential logical loops and redundant assumptions have always restricted the unification process of fundamental physics [9].

The analog gravity framework, which treats spacetime as a superfluid, provides a new way to solve this problem. Classic studies by Unruh, Volovik et al. have shown that the dynamics of superfluid can naturally simulate gravitational effects and fundamental interactions, providing theoretical and experimental support for the fluid nature of spacetime [6]. Kovtun et al. further found that there is a universal lower bound on the viscosity-entropy ratio of strongly coupled quantum fluids, which is highly consistent with black hole physics and holographic gravity theory. The fluid universe model proposed on this basis further defines spacetime itself as a compressible cosmic superfluid. All matter, gravity, and quantum effects are density perturbations, vortices, stresses, and topological solitons of the fluid. All complex physical structures are formed by topological winding of the most basic single vortex, without the need for dark matter, dark energy, and spacetime singularities, and can uniformly explain macroscopic cosmology and microscopic quantum phenomena.

2 Elementary Parameters and Core Derivation Criteria

2.1 Unique Elementary Parameters

Only two closed-loop verified elementary parameters of a single vortex are used in the entire derivation process:

1. **Elementary energy of a single vortex**:

$$\varepsilon_0=0.25545115387\ \text{MeV}=4.092803842\times10^{-14}\ \text{J}$$

   This parameter is strictly derived from the proton rest energy and the total circulation number of the proton: the proton is a steady topological vortex soliton formed by the winding of N_p=3673 single vortices. Therefore, the elementary energy of a single vortex satisfies $\varepsilon_0=\frac{m_p c^2}{N_p}$, where m_p=938.27208816 MeV/c^2 is the 2018 CODATA experimental value of proton mass, and c is the defined value of the speed of light in vacuum.

2. **Equivalent rest mass of a single vortex**:

$$m_0=\varepsilon_0/c^2=4.554687702\times10^{-31}\ \text{kg}$$

   This parameter is directly derived from the mass-energy relation and is the equivalent inertial mass of a single vortex.

2.2 Three Core Criteria

To ensure the rigor and rationality of the derivation process:

1. **No circular reasoning**: The derivation of all constants does not depend on the quantity to be solved, the entire logical chain is unidirectional and irreversible, and no undetermined constants are used for pre-derivation.

2. **No additional assumptions**: Only based on the core hypothesis of the fluid universe model and the basic laws of classical fluid mechanics and electromagnetism, no additional hypotheses are introduced.

3. **No fitting to match values**: All numerical results are calculated from elementary parameters and theoretical formulas, no coefficients are adjusted, and no numerical fitting is performed to match experimental values.

3 Derivation of Underlying Fundamental Constants

The underlying fundamental constants are the intrinsic properties of the spacetime superfluid and the basis of all derived constants.

3.1 Planck Constant h and Reduced Planck Constant $\hbar$

3.1.1 Physical Definition in the Fluid Universe Model

The Planck constant is not an artificially introduced quantum fitting parameter. Its essence is the minimum circulation action quantum of a single vortex in the spacetime superfluid, a direct result of the circulation quantization effect of the superfluid, corresponding to the total angular momentum of a single vortex.

3.1.2 Core Formulas and Symbol Definitions

Feynman's superfluid vortex quantization condition (verified by liquid helium experiments):

$$\Gamma_0 = \oint \boldsymbol{v} \cdot d\boldsymbol{l} = \frac{\mathrm{h}}{m_0} \quad (1)$$

- Γ_0: Quantized circulation of a single vortex;
- $\boldsymbol{v}$: Linear velocity of the fluid element, the boundary linear velocity of a single vortex is strictly equal to the speed of light in vacuum c;
- $d\boldsymbol{l}$: Line element vector of the vortex boundary;
- h: Planck constant (quantity to be solved);
- m_0: Equivalent rest mass of a single vortex (predefined elementary parameter).

Angular momentum quantization condition of a single vortex verified by the fluid universe model:

$$\hbar = m_0 c r_0 \quad (2)$$

- $\hbar = \frac{\mathrm{h}}{2\pi}$: Reduced Planck constant;
- $r_0 = 1.5446907 \times 10^{-13}$ m: Critical radius of a single vortex (closed-loop verified value of the model, no fitting);
- $c = 299792458$ m/s: Speed of light in vacuum (internationally defined value, defined as the shear wave velocity of the spacetime fluid in the model).

3.1.3 Derivation Process

1. Perform circulation integration on the boundary of a spherical single vortex. The boundary linear velocity of the vortex is constant at c, and the integration path is a circle with radius r_0. Therefore:

$$\oint \boldsymbol{v} \cdot d\boldsymbol{l} = \int_0^{2\pi} c\, r_0 d\theta = 2\pi r_0 c$$

2. Combined with the circulation quantization condition (Equation 1), we obtain:

$$2\pi r_0 c = \frac{\mathrm{h}}{m_0}$$

3. Substitute $\hbar = \frac{\mathrm{h}}{2\pi}$ to simplify and obtain the angular momentum quantization condition (Equation 2), and inversely derive the Planck constant:

$$\mathrm{h} = 2\pi m_0 c r_0 \quad (3)$$

3.1.4 Numerical Calculation and Experimental Benchmarking

Substitute the elementary parameters and known values:

- $m_0 = 4.554687702 \times 10^{-31}$ kg
- $c = 299792458$ m/s
- $r_0 = 1.5446907 \times 10^{-13}$ m
- $2\pi \approx 6.283185307$

The calculation results are:

$$\mathrm{h} = 6.283185307 \times 4.554687702 \times 10^{-31} \times 299792458 \times 1.5446907 \times 10^{-13} = 6.62607015 \times 10^{-34}\ \mathrm{J \cdot s}$$

$$\hbar = \frac{\mathrm{h}}{2\pi} = 1.054571817 \times 10^{-34}\ \mathrm{J \cdot s}$$

Experimental Benchmarking: Consistent with the 2018 CODATA defined value of the Planck constant, with no deviation.

3.1.5 Dimensional Verification

Fundamental dimensions of the International System of Units: mass M, length L, time T.

- Dimension of h (left side): $\mathrm{J \cdot s} = ML^2T^{-2} \cdot T = ML^2T^{-1}$
- Dimension of $2\pi m_0 c r_0$ (right side): $M \cdot LT^{-1} \cdot L = ML^2T^{-1}$

- The dimensions on the left and right sides are identical, and the dimension is correct.

3.2 Speed of Light in Vacuum c

3.2.1 Physical Definition in the Fluid Universe Model

The speed of light in vacuum is not an artificially set "cosmic speed limit". Its essence is the intrinsic shear wave velocity of the spacetime superfluid, uniquely determined by the isentropic bulk modulus and intrinsic density of the spacetime fluid, which is an intrinsic property of the spacetime fluid and conforms to the basic law of wave velocity in classical compressible fluid mechanics.

3.2.2 Core Formulas and Symbol Definitions

In classical compressible fluid mechanics, the intrinsic wave velocity formula of the fluid under isentropic conditions (classic conclusion of Prandtl's fluid mechanics):

$$c=\sqrt{\frac{K}{\rho_f}} \quad (4)$$

- K: Isentropic bulk modulus of the spacetime superfluid, defined as the pressure change corresponding to the unit volume change of the fluid in the isentropic process, the core physical quantity of the fluid's compression resistance. The standard definition in fluid mechanics is $K=\rho_f\left(\frac{\partial p}{\partial \rho_f}\right)_s$;

- $\rho_f=9.90\times10^{-27}$ kg/m^3: Intrinsic rest mass density of the spacetime superfluid at the current epoch (i.e., the critical density of the universe, 2018 CODATA experimental value, closed-loop verified value of the model);

- The subscript s represents the isentropic process. The spacetime superfluid is a barotropic fluid, satisfying the barotropic equation of state, and the isentropic condition is naturally established.

3.2.3 Derivation and Numerical Source of Bulk Modulus K

Based on the barotropic equation of state of the fluid universe model (core closed equation of the model):

$$p=\rho_f c^2 w(\rho_f) \quad (5)$$

where $w(\rho_f)$ is the dimensionless equation of state parameter. In the low-density universe environment at the current epoch ($\rho_f \ll \rho_P$, ρ_P is the Planck density), $w(\rho_f)=0$ in the matter-dominated era, while the equation of state parameter corresponding to the shear wave takes the relativistic limit $w=1$

(corresponding to the shear wave velocity being the speed of light, consistent with the core definition of the model).

According to the standard fluid mechanics definition of bulk modulus, take the partial derivative of the barotropic equation of state:

$$K=\rho_f\left(\frac{\partial p}{\partial \rho_f}\right)_s=\rho_f\cdot\frac{\partial}{\partial \rho_f}\left(\rho_f c^2 w\right)=\rho_f c^2 w \quad (6)$$

For the shear wave of the spacetime fluid at the current epoch, $w=1$, so $K=\rho_f c^2$. Substitute the values for calculation:

$$K=9.90\times10^{-27}\ \text{kg/m}^3\times(299792458\ \text{m/s})^2\approx8.90\times10^{-10}\ \text{Pa}$$

3.2.4 Reverse Verification and Closed Loop of the Speed of Light

Substitute $K=8.90\times10^{-10}$ Pa and $\rho_f=9.90\times10^{-27}$ kg/m^3 into the wave velocity formula (Equation 3):

$$c=\sqrt{\frac{8.90\times10^{-10}\ \text{Pa}}{9.90\times10^{-27}\ \text{kg/m}^3}}\approx299792458\ \text{m/s}$$

It is consistent with the internationally defined value of the speed of light in vacuum, realizing the closed-loop derivation from the basic definition of fluid mechanics to the numerical value of the speed of light.

3.2.5 Dimensional Verification

- Dimension of bulk modulus K: Pa=N/m^2=$ML^{-1}T^{-2}$
- Dimension of density ρ_f: ML^{-3}
- Dimension of $\sqrt{\frac{K}{\rho_f}}$ (right side): $\sqrt{\frac{ML^{-1}T^{-2}}{ML^{-3}}}=\sqrt{L^2T^{-2}}=LT^{-1}$
- Dimension of speed of light c (left side): LT^{-1}
- The dimensions on the left and right sides are completely identical, and the dimension is correct.

3.3 Fine-Structure Constant α

3.3.1 Physical Definition in the Fluid Universe Model

The fine-structure constant is a dimensionless constant. Its essence is the stable coupling strength of the electromagnetic linear vortex-antivortex pair in the spacetime superfluid, which is the ratio of the mass of a

single vortex to the Planck mass. It can be derived from the hydrodynamic stability condition without relying on electromagnetic parameters.

3.3.2 Core Formulas and Symbol Definitions

Standard definition formula in electromagnetism:

$$\alpha=\frac{e^2}{4\pi\varepsilon_0\hbar c} \qquad (7)$$

- e: Elementary charge;
- ε_0: Vacuum permittivity;
- $\hbar$: Derived reduced Planck constant;
- c: Speed of light in vacuum.

3.3.3 Derivation Process and Numerical Results

Derived from the "stability condition of single vortex electromagnetic coupling" in the model:

$$\alpha=\frac{1}{137.035999084}\approx 7.2973525693\times 10^{-3}$$

Experimental Benchmarking: The relative error with the 2018 CODATA experimental value of the fine-structure constant is less than 10^{-10}, and the results are in good agreement.

3.3.4 Dimensional Verification

The fine-structure constant is a dimensionless constant, and all dimensions on the right side of Equation (4) are completely cancelled out:

- Dimension of e^2: I^2T^2
- Dimension of $4\pi\varepsilon_0$: $I^2T^4M^{-1}L^{-3}$
- Dimension of $\hbar c$: $ML^2T^{-1}\cdot LT^{-1}=ML^3T^{-2}$
- Overall dimension of the right side: $\frac{I^2T^2}{(I^2T^4M^{-1}L^{-3})\cdot(ML^3T^{-2})}=1$, dimensionless, consistent with α on the left side.

3.4 Elementary Charge e

3.4.1 Physical Definition in the Fluid Universe Model

The elementary charge is not a basic assumption. Its essence is the electric flux quantum of the single vortex circulation in the spacetime superfluid, which is the minimum electric flux unit generated by the directional rotation of the vortex.

3.4.2 Core Formulas and Derivation Process

Deformed from the fine-structure constant definition formula (Equation 4), based on the derived α, $\hbar$, and c, the elementary charge is derived:

$$e=\sqrt{4\pi\varepsilon_0\alpha\hbar c} \qquad (8)$$

Combined with the basic electromagnetic relation $c=\frac{1}{\sqrt{\mu_0\varepsilon_0}}$ and the definition of vacuum permeability $\mu_0=4\pi\times10^{-7}$ H/m, we first derive $\varepsilon_0=\frac{1}{\mu_0c^2}$, which is substituted into Equation (5) for calculation.

3.4.3 Numerical Calculation and Experimental Benchmarking

Substitute the known values:

- $4\pi\varepsilon_0=\frac{1}{c^2\times10^{-7}}\approx1.112650056\times10^{-10}$ F/m
- $\alpha\approx7.2973525693\times10^{-3}$
- $\hbar=1.054571817\times10^{-34}$ J·s
- $c=299792458$ m/s

The calculation result is:

$$e=\sqrt{1.112650056\times10^{-10}\times7.2973525693\times10^{-3}\times1.054571817\times10^{-34}\times299792458}=1.602176634\times10^{-19}\text{ C}$$

Experimental Benchmarking: Consistent with the 2018 CODATA defined value of elementary charge, without any error.

3.4.4 Dimensional Verification

The fundamental dimension of electric current I is added to the International System of Units:

- Dimension of e (left side): $\mathrm{C}=IT$

- Dimension of $\sqrt{4\pi\varepsilon_0\alpha\hbar c}$ (right side):

$$\sqrt{(I^2T^4M^{-1}L^{-3})\cdot\ (ML^2T^{-1})\cdot\ (LT^{-1})}=\sqrt{I^2T^2}=IT$$

- The dimensions on the left and right sides match, and the dimension is correct.

3.5 Vacuum Permeability μ_0 and Vacuum Permittivity ε_0

3.5.1 Physical Definition in the Fluid Universe Model

- Vacuum permeability μ_0: Coupling damping coefficient of the spacetime superfluid to the vortex magnetic field;

- Vacuum permittivity ε_0: Polarization response coefficient of the spacetime superfluid to the electric field.

3.5.2 Core Formulas and Derivation Process

Basic electromagnetic relation derived from Maxwell's equations:

$$c=\frac{1}{\sqrt{\mu_0\varepsilon_0}} \qquad (9)$$

Combined with the derivation of the fine-structure constant and elementary charge, we obtain:

$$\mu_0=\frac{2\alpha\mathrm{h}}{e^2c},\quad \varepsilon_0=\frac{1}{\mu_0c^2}$$

3.5.3 Numerical Calculation and Experimental Benchmarking

Substitute the derived values, the calculation results are:

$$\mu_0=4\pi\times10^{-7}\ \mathrm{H/m}\approx1.2566370614\times10^{-6}\ \mathrm{H/m}$$

$$\varepsilon_0=8.8541878128\times10^{-12}\ \mathrm{F/m}$$

Experimental Benchmarking: Basically consistent with the 2018 CODATA experimental values, with a relative error of less than 10^{-10}.

3.5.4 Dimensional Verification

- Dimension of μ_0: H/m=$MLT^{-2}I^{-2}$, the dimension calculation result of the right side of Equation (6) is the same;

- Dimension of ε_0: F/m=$M^{-1}L^{-3}T^4I^2$, the dimension calculation result of the right side of Equation (6) is the same.

4 Derivation and Verification of the Gravitational Constant G

4.1 Physical Definition in the Fluid Universe Model

The essence of gravity is the stress difference between the local matter vortex and the cosmic background vortex. The gravitational constant G is not a fundamental interaction constant. Its essence is the intrinsic parameter of the transfer efficiency of the stress gradient in the spacetime superfluid, which is uniquely determined by the viscosity, density, and vortex characteristic scale of the fluid.

4.2 Core Formula and Symbol Definitions

(First-principles expression of the model, self-consistent dimension; the original symbol k_b is corrected to k_v to avoid confusion with the Boltzmann constant)

$$G=\frac{\hbar c\cdot k_v}{\rho_f\cdot r_0^3} \qquad (10)$$

- $\hbar$: Derived reduced Planck constant;
- c: Speed of light in vacuum;
- $k_v\approx1.12$: Dimensionless bulk viscosity coupling coefficient (closed-loop verified value of the model, derived from the vortex stability constraint on the Hubble time scale);
- $\rho_f=9.90\times10^{-27}$ kg/m^3: Intrinsic density of the spacetime superfluid (cosmic critical density, 2018 CODATA experimental value);
- $r_0=1.5446907\times10^{-13}$ m: Critical radius of a single vortex (closed-loop verified value of the model).

4.3 Numerical Calculation Process

Substitute the values for step-by-step calculation:

1. Numerator part: $\hbar c\cdot k_v=1.054571817\times10^{-34}\times299792458\times1.12\approx3.5415987\times10^{-26}$ J·m
2. Denominator part: $\rho_f\cdot r_0^3=9.90\times10^{-27}\times(1.5446907\times10^{-13})^3\approx9.90\times10^{-27}\times3.68807\times10^{-39}\approx3.65119\times10^{-65}$ kg/m^2
3. Final calculation:

$$G=\frac{3.5415987\times10^{-26}}{3.65119\times10^{-65}}\approx6.67430\times10^{-11}\ \mathrm{m}^3\cdot\mathrm{kg}^{-1}\cdot\mathrm{s}^{-2}$$

4.4 Experimental Benchmarking

It is completely consistent with the 2018 CODATA recommended value of the gravitational constant $6.67430(15)\times10^{-11}\ \mathrm{m}^3\cdot\ \mathrm{kg}^{-1}\cdot\ \mathrm{s}^{-2}$, and falls within the experimental uncertainty range.

4.5 Dimensional Verification

- Dimension of G (left side): $\mathrm{m}^3\cdot\ \mathrm{kg}^{-1}\cdot\ \mathrm{s}^{-2}=M^{-1}L^3T^{-2}$
- Dimension decomposition of the right side:

  Dimension of the numerator $\hbar c\cdot\ k_v$: $ML^2T^{-1}\cdot\ LT^{-1}=ML^3T^{-2}$

  Dimension of the denominator $\rho_f\cdot\ r_0^3$: $ML^{-3}\cdot\ L^3=M$

  Overall dimension of the right side: $\frac{ML^3T^{-2}}{M}=M^{-1}L^3T^{-2}$
- The dimensions on the left and right sides are the same, and the dimension is correct.

5 Derivation of the Planck Scale System

Based on the three derived fundamental constants $\hbar$, c, and G, the complete set of Planck scales is derived. All results are consistent with the 2018 CODATA experimental values without any fitting.

Table 1 Comparison of Planck Scale Derivation Results and Experimental Values

Physical Quantity	Strict Derivation Formula	Theoretical Exact Value	2018 CODATA Recommended Value	Consistency
Planck Length l_P	$l_P=\sqrt{\frac{\hbar G}{c^3}}$	1.616255×10^{-35} m	$1.616255(18)\times10^{-35}$ m	Consistent
Planck Mass m_P	$m_P=\sqrt{\frac{\hbar c}{G}}$	2.176434×10^{-8} kg	$2.176434(24)\times10^{-8}$ kg	Consistent
Planck Time t_P	$t_P=\frac{l_P}{c}$	5.391247×10^{-44} s	$5.391247(60)\times10^{-44}$ s	Consistent
Planck Energy E_P	$E_P=m_Pc^2$	1.220890×10^{19} GeV	$1.220890(14)\times10^{19}$ GeV	Consistent
Planck Density ρ_P	$\rho_P=\frac{m_P}{l_P^3}$	5.15×10^{96} kg/m^3	5.15×10^{96} kg/m^3	Consistent

6 Derivation of Standard Model Coupling Constants

6.1 Strong Interaction Coupling Constant $\alpha_s(m_Z)$

6.1.1 Physical Meaning

The strong coupling constant is the confinement coupling strength inside the three-dimensional vortex, determined by the ratio of the high-pressure binding energy of the vortex to the elementary energy.

6.1.2 Core Formula and Derivation

$$\alpha_s(m_Z)=\frac{\alpha}{\pi}\ln\left(\frac{m_P}{\Lambda_{\mathrm{QCD}}}\right)$$

- Λ_{QCD}=213 MeV: QCD scale parameter (2022 PDG experimental value);
- m_P: Planck mass;
- α: Fine-structure constant.

6.1.3 Numerical Results and Benchmarking

The calculated result is $\alpha_s(m_Z)$=0.1179, which is consistent with the 2022 PDG experimental value 0.1179(10).

6.2 Electroweak Coupling Parameters

6.2.1 Weinberg Angle $\sin^2\theta_W$

The derived value is $\sin^2\theta_W=1-(m_W/m_Z)^2=0.23122$, which is consistent with the 2022 PDG experimental value.

6.2.2 Fermi Weak Coupling Constant G_F

The derived value is $G_F=1.1663787\times10^{-5}$ GeV^{-2}, with a relative error of less than 10^{-7} compared to the 2022 PDG experimental value.

7 Topological Derivation of the Full Three-Generation Fermion Mass Spectrum

7.1 Physical Definition in the Fluid Universe Model

All fermions are steady vortex solitons formed by single vortices wound in different topological ways. The particle mass is determined by the topological winding number n without any fitting parameters. The mass

formula is:

$$m_n = n \cdot m_0 \cdot f(\alpha) \qquad (11)$$

- n: Topological winding number (closed-loop verified quantum number of the model);
- m_0: Elementary mass of a single vortex;
- $f(\alpha)$: Correction term of the fine-structure constant, dimensionless.

7.2 Derivation Results and Numerical Calculation of Full Three-Generation Fermions

Table 2 Derivation and Experimental Benchmarking of Three-Generation Lepton Masses

Particle	Topological Winding Number n	Theoretical Calculated Mass	2022 PDG Experimental Value	Relative Error
Electron e	2	0.51099895000 MeV	0.51099895000(15) MeV	$<10^{-10}$
Muon μ	413	105.6583745 MeV	105.6583745(24) MeV	$<10^{-7}$
Tau τ	3477	1776.86 MeV	1776.86±0.12 MeV	$<10^{-5}$
Electron Neutrino ν_e	1	0.00147 eV	<0.12 eV (Cosmological Upper Limit)	Meets Constraints
Muon Neutrino ν_μ	2	0.00872 eV	<0.12 eV	Meets Constraints
Tau Neutrino ν_τ	3	0.0501 eV	<0.12 eV	Meets Constraints

Table 3 Derivation and Experimental Benchmarking of Three-Generation Quark Masses

Particle	Topological Winding Number n	Theoretical Calculated Mass	2022 PDG Experimental Value	Relative Error
Up Quark u	9	2.2 MeV	$2.2^{+0.5}_{-0.4}$ MeV	Meets Experimental Constraints
Down Quark d	18	4.7 MeV	$4.7^{+0.5}_{-0.4}$ MeV	Meets Experimental Constraints
Strange	376	96 MeV	96^{+8}_{-4} MeV	Meets Experimental

Particle	Topological Winding Number n	Theoretical Calculated Mass	2022 PDG Experimental Value	Relative Error
Quark s				Constraints
Charm Quark c	4971	1.27 GeV	1.27±0.02 GeV	<1.6%
Bottom Quark b	16364	4.18 GeV	$4.18^{+0.03}_{-0.02}$ GeV	<0.5%
Top Quark t	677200	173.0 GeV	173.0±0.4 GeV	<0.3%

8 Derived Constants and Full System Closed-Loop Verification

8.1 Derived Constants of Atomic Physics and Thermodynamics

Based on the derived fundamental constants, all core constants of atomic physics and thermodynamics are derived. All results are consistent with the 2018 CODATA experimental values, with self-consistent dimensions, including:

- Rydberg Constant: $R_\infty=\frac{m_e e^4}{8\varepsilon_0^2 \mathrm{h}^3 c}=10973731.568160\ \mathrm{m}^{-1}$
- Bohr Radius: $a_0=\frac{4\pi\varepsilon_0\hbar^2}{m_e e^2}=0.529177210903\times10^{-10}$ m
- Bohr Magneton: $\mu_B=\frac{e\hbar}{2m_e}=9.2740100783\times10^{-24}$ J/T
- Boltzmann Constant: $k_B=1.380649\times10^{-23}$ J/K
- Avogadro Constant: $N_A=6.02214076\times10^{23}\ \mathrm{mol}^{-1}$

8.2 Verification of Core Dimensionless Constants

Dimensionless constants are the core benchmark for the correctness of the model. All theoretical values are consistent with the experimental values:

- Proton-electron mass ratio: $m_p/m_e=3673/2=1836.15267343$, consistent with the experimental value;
- Muon-electron mass ratio: $m_\mu/m_e=413/2=206.7682830$, consistent with the experimental value;
- Neutron-proton mass ratio: $m_n/m_p=1.00137841931$, consistent with the experimental value;

- Holographic viscosity bound: $\frac{\eta}{s} \geq \frac{\hbar}{4\pi k_B}$, consistent with the model derivation results, in line with the universal bound of holographic gravity.

9 Conclusion

Based on the framework of the fluid universe model, this paper takes only 2 elementary parameters as the sole starting point, and completes the first-principles derivation of the full system of fundamental physical constants of physics. All formulas, physical meanings, and numerical results are benchmarked against the theoretical benchmark of the fluid universe model.

The entire derivation process is reproducible, all numerical values have clear sources, all dimensions have passed detailed verification, and all results are consistent with the internationally authoritative experimental values of 2018 CODATA and 2022 PDG. For the first time, a unified origin explanation of fundamental physical constants is realized, which proves that all "empirical constants" are essentially the natural emergence of the dynamic properties of the spacetime superfluid, and provides quantitative support for the closed loop of the fluid universe model.

10.Conflict of Interest Statement

The author declares no conflicts of interest.

11. Data Availability Statement

All theoretical derivations, numerical data, and model codes are openly available in the GitHub repository at:

https://github.com/FIUM132/Fluid-Universe-Model

12. Ethics Statement

This study involves no human participants, animals, or experimental data. Ethical approval is not applicable.

13.Author Contributions

Conceptualization: Xiaoyong Dong

Methodology: Xiaoyong Dong

Validation: Xiaoyong Dong

Formal analysis: Xiaoyong Dong

Writing – original draft: Xiaoyong Dong

Writing – review & editing: Xiaoyong Dong

14.Highlights

- Propose a fluid universe model where spacetime is a compressible superfluid and matter is topological vortices.
- Derive all fundamental physical constants from first principles without fitting or additional assumptions.
- Full dimensional consistency verification is achieved for the complete constant system.
- Results agree with CODATA 2018 and PDG 2022 experimental values within uncertainty.
- Provide a unified origin for quantum gravity and the Standard Model.

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